

Event-triggered adaptation of central pattern generator parameters in quadrupedal robot locomotion

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Abstract—Central pattern generators (CPGs) produce efficient rhythmic locomotion from a small set of parameters, but a parameter set tuned for one terrain is no longer optimal once the terrain changes. We show that the CPG parameters yielding stable walking on flat ground and on a 10° incline are distinct, and that on a corridor of repeated transitions keeping the flat-optimal set more than doubles the fall rate relative to the terrain-matched optimum, a gap that persists over the traverse. Resuming stable locomotion therefore requires online adaptation. We propose an event-triggered scheme in which a prediction-error monitor detects each terrain change and retunes the CPG within the same walking bout, and we compare grid search, Bayesian optimization, and an autoregressive active inference agent that minimizes expected free energy. Grid search and Bayesian optimization destabilize the robot while probing candidate gaits, roughly doubling the fall rate, whereas the active inference agent selects parameters from a learned model without such trials and avoids this harm. Aligning its goal prior with the stability criterion, rather than with velocity tracking, makes it significantly outperform keeping the incumbent gait on a sustained transition, and the information-seeking term of the free energy is essential to this gain; generalizing the improvement across repeated transitions remains open. Selecting parameters from a model avoids the destabilization of trial-based search, and encoding stability in the agent’s goal is what turns caution into improvement.

I. INTRODUCTION

Achieving the same level of adaptability as animals is crucial for deploying legged machines in constantly changing real world environments. To bridge the gap between engineering and nature, engineers have frequently turned to biology for ideas, replicating animal morphologies. For example, snake-like robots for constrained environments or quadruped platforms for irregular terrain [8], [3]. These morphologies inherit the mechanical advantages of their biological counterparts. However, their control systems often lack in adaptability. For example, the Boston Dynamics quadrupedal robot, Spot, mimics animal legs. However, it relies on centralized and computationally intensive controllers that require multiple sensors [3]. Although this approach performs well, it remains complex and expensive, limiting its scalability for wider deployment and access.

To address these challenges, a promising approach to improve robotic locomotion is to use Central Pattern Generators (CPGs), which are neural circuits that generate high-dimensional rhythmic signals. In vertebrates, CPGs are located in the spinal cord and produce periodic coordinated

movements such as walking and breathing [5]. By automating these repetitive physical patterns, CPGs allow the brain to redirect its resources to other cognitive tasks, such as selecting safe paths or responding to unexpected stimuli. These systems are triggered by simple inputs from higher brain centers. The brain can modulate their rhythmic output to control parameters such as locomotion speed and gait. Moreover, CPGs remain coupled to the body and environment through sensory feedback [2]. This decentralized architecture reduces the delays for commands and the required bandwidth while simplifying high-level control [6]. These features make CPG-based controllers an attractive solution as they are less energy intensive, simpler, and more affordable.

A. Related Work

Inspired by biological principles, various CPG models have been implemented in robotics. Among them, coupled nonlinear oscillators have emerged as biologically plausible and computationally efficient for generating rhythmic movement [1], [6].

The power of coupled nonlinear oscillators lies in their ability to inhibit each other. This coupling behavior enables synchronization between joints, producing coordinated gaits and smooth transitions between them [1], [11], [9]. Such mechanisms mimic biological inter-neuronal inhibition and allow the generation of out-of-phase rhythmic signals. They also exhibit a limit-cycle behavior, producing periodic and self-sustaining oscillations that quickly recover from transient perturbations [11]. In addition, their compact dynamics, governed by a small set of parameters (e.g. frequency and amplitude), makes them appropriate for embedded systems with limited resources. Despite their promise, most existing implementations target high-end robotic platforms. Their applicability to compact, affordable systems remains under-explored, which underpins the present study.

A complementary line of work treats gait tuning as black-box optimization, adjusting the controller parameters from measured performance rather than from a model of the dynamics. Bayesian optimization is a common choice when each evaluation is expensive, as a physical or simulated walking trial is. It fits a probabilistic surrogate, typically a Gaussian process, to the parameter-score pairs observed so far, and selects the next parameters by maximizing an acquisition function such as the upper confidence bound, which favors settings that are either predicted to score highly or are still uncertain. Run *online*, the procedure scores the parameters used over the most recent time window, augments the surrogate with that observation, and proposes

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the parameters for the next window, so the gait is refined during walking rather than between episodes. Because unconstrained proposals can jump to destabilizing gaits, the search is confined to a trust region around the current best parameters [4]. We adopt this online Bayesian optimization as one of the adaptation baselines in Section IV.

In the following, we contribute

- evidence that terrain-matched CPG parameters confer a durable survival advantage across repeated terrain transitions, so that resuming stable locomotion calls for online adaptation;
- an event-triggered adaptation scheme, driven by a re-arming prediction-error monitor, together with the finding that online black-box gait optimization (grid search and Bayesian optimization) is actively destructive on hazardous transitions; and
- an autoregressive active inference agent that avoids this destabilization by selecting from a learned model, and the result that aligning its goal prior with the stability criterion is what turns this caution into a significant improvement over the incumbent gait on a sustained transition, with robust adaptation across recurring transitions identified as an open problem.

II. PROBLEM STATEMENT

We first introduce the central pattern generator that produces coordinated leg motion in a quadrupedal robot. We then argue that the parameters yielding stable locomotion depend on the terrain, so that a parameter set tuned for one terrain degrades once the terrain changes. This motivates online adaptation of the parameters as a requirement for resuming stable locomotion.

A. Central Pattern Generator

The CPGs are modeled as a network of four mutually coupled Hopf oscillators, each driving one limb: left front (LF), right front (RF), left hind (LH), and right hind (RH). Coordination between limbs arises from inhibitory coupling through a gait-specific coupling matrix $K = [k_{ij}]$. Each oscillator evolves as:

$$\dot{x}_i = \alpha(u - r_i^2)x_i - \omega_i y_i, \quad (1)$$

$$\dot{y}_i = \beta(u - r_i^2)y_i + \omega_i x_i + \gamma \sum_{j=1}^4 k_{ij} y_j + s_i, \quad (2)$$

$$\omega_i = \frac{\omega_{\text{stance}}}{1 + e^{-b y_i}} + \frac{\omega_{\text{swing}}}{1 + e^{b y_i}}, \quad (3)$$

where $r_i = \sqrt{x_i^2 + y_i^2}$, $u = 2$ is the fixed squared limit-cycle amplitude (a constant, not tuned), $\alpha, \beta > 0$ are convergence rates, γ is the coupling gain, and b controls the sharpness of the frequency transition between stance and swing phases.

The final term s_i in Eq. 2 is the contact-driven sensory feedback, and it is where the fast-contact gain F_{fast} and the stop gain K_{stop} enter the dynamics. It couples each oscillator to its leg's stance or swing phase, read from y_i by hysteresis,

and to the binary foot-contact signal $c_i \in \{0, 1\}$:

$$s_i = \begin{cases} K_{\text{stop}}(\omega_i x_i - \gamma \sum_j k_{ij} y_j), & \text{stop,} \\ F_{\text{fast}} \text{sgn}(y_i), & \text{fast,} \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

The *stop* case applies when phase and contact disagree, that is a swing phase with no contact ($c_i = 0$) or a stance phase with contact ($c_i = 1$), and K_{stop} scales a correction that arrests the oscillator; the *fast* case applies otherwise during swing or stance, where F_{fast} advances it in the direction of y_i ; the feedback is zero during the transition between phases. This is a Righetti-style load feedback that keeps the rhythm entrained to ground contact [9].

The CPG outputs (x_i, y_i) are mapped to hip and knee joint angles:

$$\phi_{\text{hip}} = \phi_{\text{hip}}^0 + A_{\text{hip}} x_i, \quad (5)$$

$$\phi_{\text{knee}} = \phi_{\text{knee}}^0 - A_{\text{knee}} \max(0, y_i), \quad (6)$$

where x_i drives the forward-backward hip swing and y_i controls knee flexion, the fixed offsets ϕ_{hip}^0 and ϕ_{knee}^0 set the neutral pose, the gains A_{hip} and A_{knee} scale the swing and flexion amplitudes, and negative y_i is suppressed to prevent hyperextension.

The behavior of the CPG is governed by a small set of parameters, collected in the vector

$$\theta = [\gamma, \omega_{\text{swing}}, \omega_{\text{stance}}, F_{\text{fast}}, K_{\text{stop}}, A_{\text{hip}}, A_{\text{knee}}, b], \quad (7)$$

comprising the coupling gain, the phase-dependent stance and swing frequencies, the fast-contact gain F_{fast} and stop gain K_{stop} of the sensory feedback (Eq. 4), the hip and knee amplitudes, and the sigmoid sharpness b of Eq. 3. These parameters set the gait shape, and are held fixed during locomotion.

B. Terrain-dependent parameters

For a fixed terrain, the parameter vector θ can be tuned to produce stable locomotion. During a bout the robot reports its forward velocity v_x and the roll and pitch of its trunk, ψ_r and ψ_p , which measure the deviation from an upright attitude. Let e denote the terrain. We score a parameter set over an evaluation window by

$$V(\theta, e) = \min\left(\frac{\bar{v}_x}{v^*}, 1\right) - \frac{1}{\psi_0} \left(\text{RMS}[\psi_r] + \text{RMS}[\psi_p] \right), \quad (8)$$

and set $V = -c_f$ whenever the robot falls. Here $\bar{v}_x$ is the mean forward velocity, v^* the target speed, and ψ_0 normalizes the attitude penalty; the root-mean-square is taken over the window on low-pass-filtered angles, so that the gait's rhythmic rocking is not penalized, and $\tilde{\psi}_p$ is the pitch measured relative to the terrain incline. The progress term saturates at v^* , so moving slower up an incline or faster down it carries no penalty as long as the robot keeps advancing toward its goal, while the attitude term dominates the score, encoding that stable locomotion is the primary requirement.

The best gait for terrain e is then the terrain-conditioned optimum

$$\theta^*(e) = \arg \max_{\theta} V(\theta, e). \quad (9)$$

The difficulty is that $\theta^*(e)$ is not constant across terrains. A gait that keeps the trunk level on flat ground pitches the body, shifts foot-contact timing, and alters the coupling needed for coordination once the same legs meet an incline, so the location of the optimum moves. We consider the two terrains of Fig. 1: flat ground and a 10° incline. Optimizing Eq. 8 on each terrain with Bayesian optimization yields the parameter vectors shown in the lower panel, and they are visibly different. The incline optimum uses markedly stronger inter-limb coupling, with the coupling gain γ raised from the bottom to the top of its range, together with a much softer stance-to-swing frequency transition, expressed by a small sharpness b ; the swing frequency ω_{sw} and stop gain K_{stop} shift only slightly, and the remaining four parameters are identical across the two terrains. Each set is more stable on the terrain it was tuned for. The cost of the mismatch grows with exposure to the new terrain. On a single, isolated transition onto the incline it is transient: retaining the flat optimum roughly doubles the early fall rate, but survivors eventually settle and the advantage of the incline optimum washes out over the bout (Section IV). On a corridor of repeated 10° transitions, however, the cost is sustained: keeping the flat optimum leads to a fall in 26% of runs, against 11% when the parameters are switched to the incline optimum, and this gap holds over the whole traverse rather than fading (Section IV). Terrain-matched parameters therefore confer a durable survival advantage whenever the robot must cross terrain changes repeatedly, which is the setting a legged machine actually operates in.

Because θ is held fixed, a change in terrain leaves the controller operating at a now-suboptimal point $\theta^*(e_{old}) \neq \theta^*(e_{new})$. The resulting attitude deviations compound over time: as the trunk tilts away from balance, recovery demands larger corrections, and locomotion can fail before the robot has traversed the new terrain. Fig. 2 shows a representative bout. While the terrain is flat the trunk stays within a couple of degrees of vertical, but when the robot meets the incline with the flat-optimal parameters unchanged, its tilt grows over a few seconds until it topples and forward progress stops. Switching to the incline-optimal parameters at the transition instead holds the tilt near the modest baseline set by the incline, and the robot keeps climbing. To resume stable locomotion after a terrain change, the CPG parameters must therefore be adapted online, which requires both detecting that the terrain has changed and re-tuning θ toward $\theta^*(e_{new})$ quickly enough to preserve balance.

III. METHODS

A. Autoregressive Active Inference Agent

The agent is based on a multivariate autoregressive model with exogenous inputs (MARX) [7]. It maintains a belief over dynamics parameters updated from observations, and

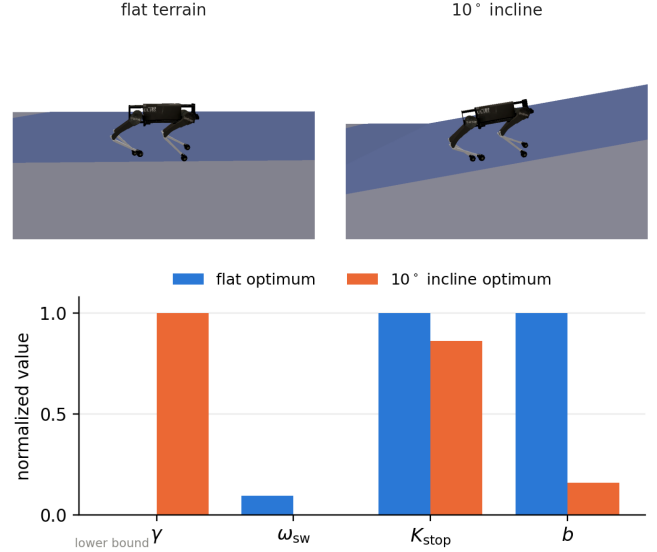


Fig. 1. The optimal CPG parameters depend on the terrain. *Top*: the Laikago walking on flat ground and on a 10° incline. *Bottom*: the parameters that differ between the two terrain-conditioned optima of Eq. 8, normalized to the shared bounds; the remaining four (ω_{st} , F_{fast} , A_{hip} , A_{knee}) are identical across the two terrains and are omitted. The shift is largest in the coupling gain γ and the frequency-transition sharpness b .

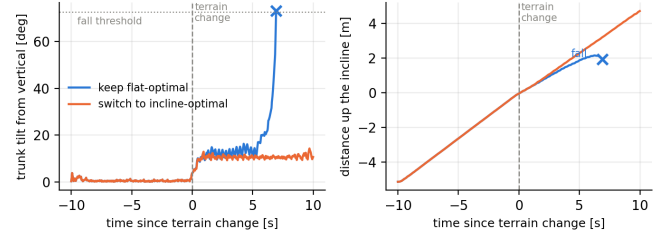


Fig. 2. Keeping the flat-optimal parameters after a terrain change destabilizes the robot. *Left*: the trunk tilt from vertical stays small on flat ground; after the transition (dashed line) it settles near the incline baseline when the parameters are switched to the incline optimum, but grows past the fall threshold when they are kept. *Right*: the robot climbs steadily after switching, whereas keeping the flat-optimal parameters ends in a fall and halted progress. Curves are one representative seed.

selects controls by minimising an Expected Free Energy (EFE) objective over a planning horizon.

1) *Model*: The likelihood is a Gaussian autoregression with coefficient matrix $A \in \mathbb{R}^{D_x \times D_y}$ and noise precision $W \in \mathbb{R}_+^{D_y \times D_y}$. Denoting $\Theta = (A, W)$:

$$p(y_k | \Theta, u_k, \bar{u}_k, \bar{y}_k) = \mathcal{N}(y_k | A^\top x_k, W^{-1}), \quad (10)$$

where $\bar{u}_k$ and $\bar{y}_k$ are running memories of the M_u previous inputs and M_y previous outputs, and the regressor concatenates the current control with the memories:

$$x_k = \begin{bmatrix} u_k \\ \bar{u}_k \\ \bar{y}_k \end{bmatrix}, \quad D_x = D_u(M_u + 1) + D_y M_y. \quad (11)$$

The prior over parameters is a matrix-normal-Wishart (MNW) distribution [10, ID: P323]:

$$p(\Theta) = \mathcal{M}\mathcal{N}(A | M_0, \Lambda_0^{-1}, W^{-1}) \mathcal{W}(W | \Omega_0^{-1}, v_0). \quad (12)$$

Controls are given an independent Gaussian prior for regularisation, with precision Υ :

$$p(u_k) = \mathcal{N}(u_k | 0, \Upsilon^{-1}). \quad (13)$$

The target observation is encoded as a Gaussian goal prior:

$$p(y_t | y_*) = \mathcal{N}(y_t | m_*, S_*). \quad (14)$$

2) *Parameter Estimation*: Because the MNW distribution is conjugate to the Gaussian likelihood, Bayesian filtering yields a closed-form MNW posterior $p(\Theta | \mathcal{D}_k) = \mathcal{MNW}(A, W | M_k, \Lambda_k^{-1}, \Omega_k^{-1}, v_k)$ at each step k , where $\mathcal{D}_k = \{y_i, u_i\}_{i=1}^k$. The parameters are updated by rank-one corrections:

$$v_k = v_{k-1} + 1, \quad (15)$$

$$\Lambda_k = \Lambda_{k-1} + x_k x_k^\top, \quad (16)$$

$$M_k = (\Lambda_{k-1} + x_k x_k^\top)^{-1} (\Lambda_{k-1} M_{k-1} + x_k y_k^\top), \quad (17)$$

$$\Omega_k = \Omega_{k-1} + y_k y_k^\top + M_{k-1}^\top \Lambda_{k-1} M_{k-1} - (\Lambda_{k-1} M_{k-1} + x_k y_k^\top)^\top (\Lambda_{k-1} + x_k x_k^\top)^{-1} (\Lambda_{k-1} M_{k-1} + x_k y_k^\top).$$

Marginalising the likelihood over the MNW prior yields a multivariate Student-t posterior predictive:

$$p(y_k | u_k, \mathcal{D}_{k-1}) = \mathcal{T}_{\eta_k}(y_k | \mu_k(u_k), \Sigma_k(u_k)), \quad (18)$$

with $\eta_k = v_{k-1} - D_y + 1$ and

$$\mu_k(u_k) = M_{k-1}^\top x_k, \quad (19)$$

$$\Sigma_k(u_k) = \frac{\Omega_{k-1}}{v_{k-1} - D_y + 1} \left(1 + x_k^\top \Lambda_{k-1}^{-1} x_k \right). \quad (20)$$

Both the mean μ_k and scale Σ_k are smooth functions of u_k .

3) *Planning*: To select controls over a horizon $\tau = (k+1, \dots, k+T)$, the agent unrolls the autoregressive model forward. At each step $t \in \tau$, the regressor is formed as $x_t = [u_t, \bar{u}_t, \hat{y}_{t-1}]^\top$, where future outputs not yet observed are substituted with MAP estimates of earlier predictives, $\hat{y}_{t-1} = [\mu_{t-1}(u_{t-1}), \mu_{t-2}(u_{t-2}), \dots]$. This yields a sequence of Student-t predictives $p(y_t | u_t, \mathcal{D}_k)$ with means $\mu_t(u_t)$ and scales $\Sigma_t(u_t)$.

The planning objective is the Expected Free Energy (EFE), defined as the sum of KL divergences between each posterior predictive and the goal prior:

$$\mathcal{J}(u_\tau) = \sum_{t \in \tau} D_{\text{KL}}[p(y_t | u_t, \mathcal{D}_k) \parallel p(y_t | y_*)]. \quad (21)$$

Expanding this KL divergence yields goal-seeking terms (functions of μ_t , penalising deviation from m_*) and information-seeking terms (functions of Σ_t , penalising output uncertainty). Together with the control prior, these produce the MAP action selection problem over the planning horizon:

$$\hat{u}_\tau = \arg \min_{u_\tau \in \mathbb{U}^T} \sum_{t \in \tau} \left[-\frac{1}{2} \ln |\Sigma_t(u_t)| + \frac{1}{2} \text{tr} \left[S_*^{-1} \frac{\eta_t}{\eta_t - 2} \Sigma_t(u_t) \right] + \frac{1}{2} (\mu_t(u_t) - m_*)^\top S_*^{-1} (\mu_t(u_t) - m_*) + u_t^\top \Upsilon u_t \right], \quad (22)$$

where $\mathbb{U}^T = \mathbb{U} \times \dots \times \mathbb{U}$ is the space of affordable controls over the horizon.

IV. EXPERIMENTS

We test whether *event-triggered* online adaptation of the CPG parameters can restore stable locomotion after a terrain change, and we compare three parameter-selection strategies as the adaptation mechanism: a *grid search* that samples the parameter space by Latin-hypercube design, online *Bayesian optimization* (BO) with a Gaussian-process surrogate and a UCB acquisition function, and the proposed *MARX-EFE* agent of Section III. Grid search and BO are black-box optimizers that can only assess a candidate gait by walking on it, whereas MARX-EFE selects parameters from its continually updated model without such trials. The comparison targets the transient after the terrain change: the hypothesis is that the method which finds better parameters faster keeps the robot upright, because a growing attitude deviation is progressively harder to correct.

A. Setup

Experiments use the Laikago quadruped in PyBullet at a control rate of 100 Hz ($\Delta t = 0.01$ s). The robot walks along a corridor that is flat for the first stretch and then rises at a 10° incline. It starts with the flat-optimal parameters $\theta^*(\text{flat})$ and, after 10 s of flat walking, meets the incline; the incline is placed at the robot's forward position at that instant, measured per seed by a calibration run, so that every run enters the slope in the same phase of its gait. The reference optima $\theta^*(\text{flat})$ and $\theta^*(10^\circ)$ are obtained beforehand by maximizing the criterion of Eq. 8 on each terrain with Bayesian optimization, using $v^* = 0.5$ m/s, $\psi_0 = 10^\circ$, and a fall penalty $c_f = 2$.

We run $N = 100$ seeds. Across-seed variability comes only from a small perturbation of the initial joint angles ($\sigma = 0.002$ rad), which the contact dynamics amplify into distinct trajectories; the start pose and the terrain geometry are otherwise identical across seeds, which isolates the response to the terrain change from the choice of starting point.

B. Detecting the terrain change

Adaptation is triggered by a prediction-error monitor. A MARX model (Eq. 10) is updated every control step from the observation $y_k = (v_x, v_y, \psi_p, \psi_r)$ and predicts the observation 0.5 s ahead under the current parameters. On flat ground this rollout error is small; when the robot meets the incline the prediction breaks down and the error rises. We compare an exponential moving average of the error against a baseline collected during flat walking, and fire the trigger when it exceeds baseline mean + $K \cdot$ baseline s.d. One-step innovations at 100 Hz are dominated by persistence and barely register the terrain change, whereas the 0.5 s rollout compounds the model mismatch into a clear spike. Calibrating on held-out seeds, $K = 16$ is the smallest threshold with no false positives on flat-only runs and no missed detections on flat-to-incline runs; it fires on average 0.33 s after the trunk crosses onto the incline. For the repeated-transition corridor of Section IV-E the monitor re-arms on the rising edge of the error, so it fires once per ramp rather than latching; a threshold of $K = 10$

TABLE I

EVENT-TRIGGERED ADAPTATION ON THE FLAT-TO- 10° TRANSITION ($N = 100$ SEEDS). LOWER IS BETTER EXCEPT UPRIGHT TIME AND $\bar{V}$. BEST ONLINE METHOD IN BOLD.

Metric	reference		online method		
	no adapt	oracle	grid	BO	MARX-EFE
Falls, 20 s [%]	40	47	83	90	39
Falls, 10 s [%]	19	8	37	37	10
Upright time [s]	16.7	17.3	12.3	12.2	17.8
Mean $\bar{V}$	-0.36	-0.30	-0.54	-0.54	-0.33

then detects every ramp with no false positives on the flat landings between them.

C. Adaptation protocol

When the trigger fires, the selected method begins proposing CPG parameters, one candidate per 1.5 s window, ramped in over 0.3 s, and each window is scored by Eq. 8. The optimizers search a trust region around the incumbent and share a common safety layer: the best-scoring parameters seen so far are restored for one window whenever a candidate scores well below them. Without the safeguard, every method, MARX-EFE included, destabilizes the robot within seconds, so it is applied uniformly across methods. The trust-region radius, however, interacts with θ^* : at ± 0.25 of the parameter range the incline optimum is not reachable, because $\theta^*(10^\circ)$ raises the coupling gain γ across its full range and drops the sharpness b , so only 30% of the flat-to-incline shift lies inside the box. We therefore report both the tight ± 0.25 region, which equalizes caution across the black-box optimizers, and a full-range region in which $\theta^*(10^\circ)$ is reachable, which is required to ask whether a method can adapt toward it. MARX-EFE uses a forgetting factor of 0.995 so that its model tracks the change, and its control prior is centered on the current parameters. Its goal prior encodes the target observation; we compare the original prior, which rewards the target forward velocity with a loose attitude term, against a *stability-aligned* prior whose attitude term is tightened to the $\psi_0 = 10^\circ$ scale of the criterion, so that the expected-free-energy objective targets the attitude that θ^* optimizes.

All arms share an identical pre-trigger trajectory per seed so that the event fires at the same instant in every arm. Two references bracket the online methods: *no adaptation* keeps $\theta^*(\text{flat})$, a lower reference, and the *oracle* switches directly to $\theta^*(10^\circ)$, an upper reference that presumes the new optimum is known. The online methods are *grid*, *BO*, and the proposed *MARX-EFE*, the last with either the velocity or the stability-aligned goal prior; we write *MARX-EFE-S* for the stability-aligned variant. We report the fall rate over the 20 s after the trigger, the fall rate over the first 10 s (the transition phase, where the parameter mismatch itself is the hazard), the mean time the robot stays upright, and the mean windowed criterion $\bar{V}$.

TABLE II

ALIGNING THE GOAL PRIOR WITH STABILITY ON THE FLAT-TO- 10° TRANSITION ($N = 100$ SEEDS, TRUST REGION WIDENED SO $\theta^*(10^\circ)$ IS REACHABLE). MARX-EFE USES THE VELOCITY GOAL PRIOR; MARX-EFE-S TIGHTENS THE ATTITUDE TERM TO THE STABILITY SCALE; THE LAST COLUMN REMOVES THE EPISTEMIC TERM FROM MARX-EFE-S. LOWER IS BETTER EXCEPT UPRIGHT TIME AND $\bar{V}$.

Metric	reference		MARX-EFE goal prior		
	no adapt	oracle	velocity	stability	stab., no epist.
Falls, 20 s [%]	40	47	44	27	42
Falls, 10 s [%]	19	8	17	7	17
Upright time [s]	16.7	17.3	16.8	18.4	17.0
Mean $\bar{V}$	-0.36	-0.30	-0.37	-0.29	-0.37

D. Results

Table I and Fig. 3 summarize the single-transition comparison at the ± 0.25 trust region. The shift in the optimum matters at the transition: within the first 10 s the oracle falls in 8% of seeds against 19% for no adaptation (McNemar $p = 0.03$), so switching to terrain-matched parameters roughly halves the early falls.

Trial-based search is destructive when run online. Grid and BO must spend whole windows walking on speculative candidates, and on an incline a poor candidate does not merely score badly, it tips the robot over. Even confined to the trust region and protected by the safeguard, this roughly doubles the fall rate relative to not adapting (83% and 90% against 40%; McNemar $p \leq 4 \times 10^{-10}$, upright-time Wilcoxon $p \approx 10^{-12}$). MARX-EFE avoids these physical trials by selecting from its model, and stays far below the black-box optimizers (early 10% against 37%, $p < 10^{-5}$). Over the full horizon, however, it matches no adaptation (39% against 40%, not significant): with the velocity goal prior it avoids harm but does not improve on holding the incumbent.

The parameter trajectories explain why (Fig. 4). Under the velocity goal prior the selected $\theta(t)$ stays at $\theta^*(\text{flat})$: MARX-EFE does not move toward the incline optimum. Two factors are responsible. The tight trust region does not contain $\theta^*(10^\circ)$, so the incline optimum is unreachable; but widening it does not help either, and the deeper cause is the goal prior, which rewarded forward velocity with only a loose (45°) attitude term and so did not target the attitude stability that θ^* optimizes. Ablations confirm that the machinery was idle under this goal: removing the epistemic term (a greedy predictive selector), removing the forgetting factor, or removing the safeguard each leaves the outcome statistically unchanged, and re-centering the control prior away from the incumbent only destabilizes the robot. The safeguard is thus load-bearing for grid and BO, which wander, but not for MARX-EFE, which holds the incumbent regardless.

Aligning the goal prior with the stability criterion changes this. Table II reports the effect of tightening the attitude term to the $\psi_0 = 10^\circ$ scale (MARX-EFE-S), with the trust region widened so that $\theta^*(10^\circ)$ is reachable. MARX-EFE-S falls

Event-triggered CPG adaptation on flat→10° slope (100 seeds, adaptation window 20 s)

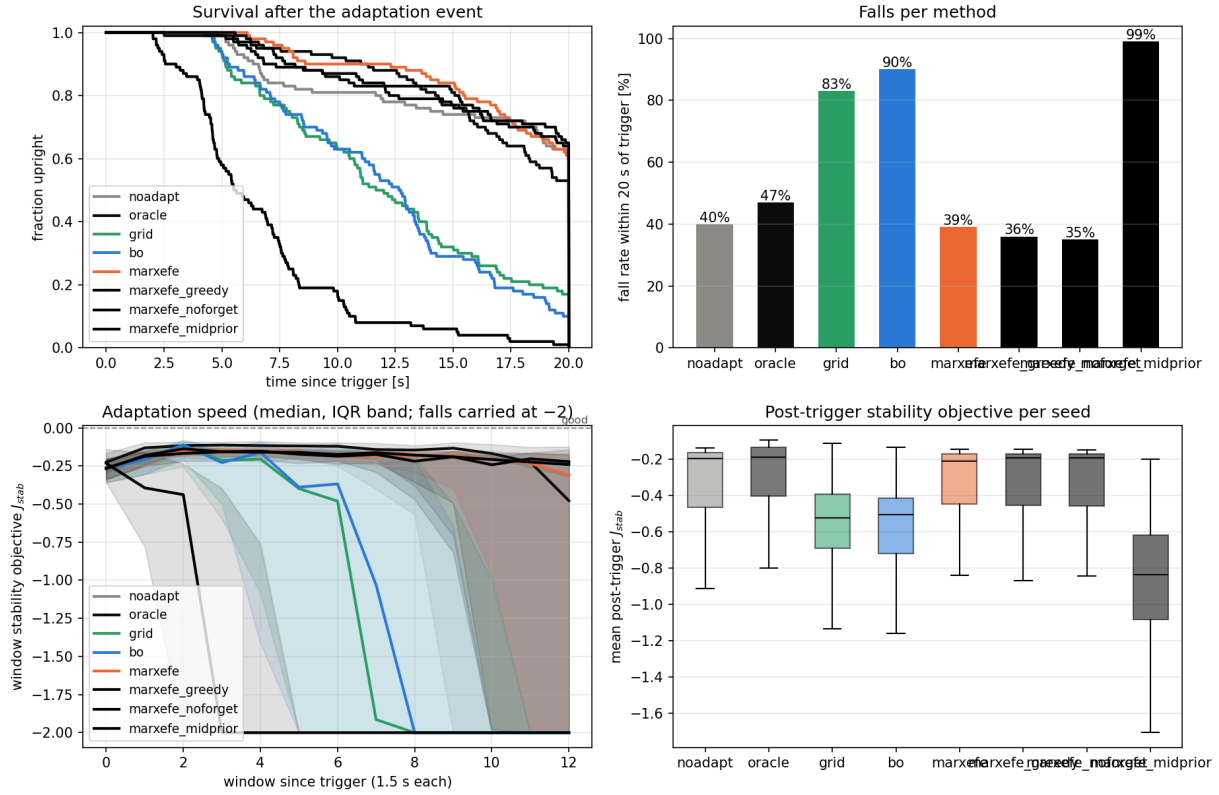


Fig. 3. Event-triggered adaptation on the flat-to-10° transition (100 seeds). *Top left*: fraction of seeds still upright versus time since the trigger. *Top right*: fall rate within 20 s. *Bottom left*: median windowed criterion V after the trigger (shaded band: interquartile range; falls carried at -2). *Bottom right*: distribution of the per-seed mean V . MARX-EFE (orange) tracks the oracle (black), while grid (green) and BO (blue) fall far more often.

in 27% of runs over 20s against 40% for no adaptation (McNemar $p = 0.04$) and in 7% against 19% over the first 10 s ($p = 0.008$), the first arm to improve significantly on holding the incumbent, and below the oracle as well ($p = 0.006$). It achieves this not by reaching $\theta^*(10^\circ)$, which stays out of range of its small adjustments, but by making minor attitude-directed corrections around the incumbent: a nearby, model-guided change suffices to keep the trunk upright. The epistemic term is now load-bearing: with it removed the same goal prior yields 42%, no better than no adaptation. The active-inference machinery therefore contributes, but only once its goal prior encodes the quantity being optimized.

E. Repeated transitions

A single incline is a mild test of adaptation, because a fixed gait that survives the transition can coast through the rest of the bout. We therefore repeat the experiment on a corridor of two consecutive 10° ramps separated by flat landings, so that the robot meets the terrain change more than once and a mismatched gait is punished repeatedly. Here the value of terrain-matched parameters is both larger and durable: the oracle falls in 11% of runs against 26% for no adaptation, and the gap holds over the whole traverse rather than washing out as it did on the single incline. The prediction-error monitor

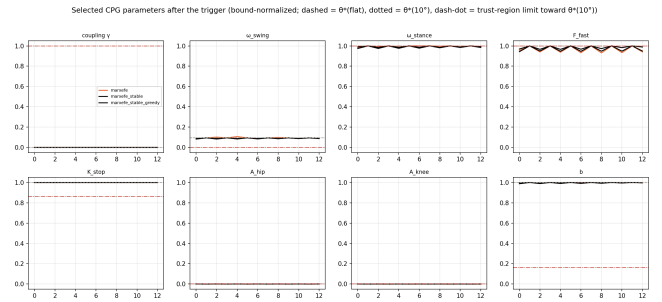


Fig. 4. Selected CPG parameters after the trigger on the flat-to-10° transition, normalized to the parameter bounds (median over 100 seeds). Dashed lines mark the flat incumbent $\theta^*(flat)$ and dotted lines the incline optimum $\theta^*(10^\circ)$. Neither the velocity-goal MARX-EFE nor the stability-aligned MARX-EFE-S travels to $\theta^*(10^\circ)$; MARX-EFE-S instead makes small attitude-directed corrections around the incumbent, which are sufficient to reduce falls (Table II).

re-arms after each ramp and detects every transition without firing on the flat landings between them. Online black-box search remains destructive: grid and BO fall in 87% and 92% of runs at the full-range trust region (38% and 41% even when confined to the tight region), far above both references.

The model-based agent does not capture the oracle’s

TABLE III
 REPEATED 10° TRANSITIONS (TWO RAMPS, $N = 100$ SEEDS,
 FULL-RANGE TRUST REGION). MARX-EFE-S IS THE
 STABILITY-ALIGNED AGENT. LOWER IS BETTER EXCEPT UPRIGHT TIME
 AND $\bar{V}$.

Metric	reference		online method		
	no adapt	oracle	grid	BO	MARX-EFE
Falls, 20 s [%]	26	11	87	92	23
Falls, 10 s [%]	9	10	65	73	11
Upright time [s]	17.4	18.7	9.6	9.0	17.7
Mean $\bar{V}$	0.22	0.44	-0.46	-0.53	0.27

advantage here, however (Table III). With either goal prior it falls in 23–24% of runs, statistically indistinguishable from no adaptation (26%, McNemar $p = 0.83$) and well above the oracle (11%, $p = 0.007$). The stability-aligned prior that improved on the single sustained incline does not replicate that gain when the transitions recur, and the epistemic term confers no advantage in this setting either. A gait corrected for one ramp is not necessarily right for the next after an intervening flat stretch, and the small corrections that sufficed on a single incline do not reach the alternating optimum. Capturing the oracle’s durable benefit across repeated transitions, rather than only avoiding the destabilization that black-box search causes, therefore remains open.

V. DISCUSSION

Our results separate two aspects of online adaptation that are easily conflated: deciding *when* to change the gait and deciding *what* to change it to. The prediction-error trigger resolves the first cheaply and reliably, firing within a third of a second of each terrain change and never on flat ground, and it re-arms to detect a sequence of transitions. The second is where the methods diverge. Grid search and Bayesian optimization can only judge a candidate gait by walking on it, and on an incline a poor candidate does not merely score badly, it tips the robot over; even confined to a trust region and protected by a revert-to-best safeguard, this exploration roughly doubles the fall rate relative to not adapting at all. The active inference agent instead selects parameters from a model that it updates every control step, so it avoids these destabilizing trials. What the agent selects, however, depends critically on its goal prior. With a goal that tracks forward velocity, the agent simply holds the incumbent gait: it avoids harm but does not improve on not adapting, and its information-seeking term, forgetting factor, and revert-to-best safeguard are all idle. Once the goal prior encodes attitude stability, matching the criterion that defines the terrain optimum, the agent instead makes small model-guided corrections that significantly reduce falls on a sustained transition, and the information-seeking term becomes essential to that gain. The advantage is therefore not faster search, but the ability to improve the gait without risking it, and it appears only when the agent’s objective encodes the quantity being optimized.

Several limitations qualify these conclusions. The improvement does not yet generalize: on a corridor of repeated transitions, where the terrain-matched optimum confers a large and durable advantage, neither the stability-aligned agent nor any goal-prior setting we swept captures it, and the selected parameters never travel to the terrain optimum. The barrier appears structural rather than a matter of tuning: the linear autoregressive model supports small local corrections but cannot predict which distant parameter settings are stable, and loosening the control prior to permit larger moves only reintroduces destabilizing exploration. Robust adaptation across recurring terrain changes, for instance by re-selecting per transition or by a model expressive enough to navigate to the terrain optimum, is the central open problem. The study is also in simulation with reference optima computed offline, and the open-loop CPG has a narrow terrain envelope: beyond roughly 10° of incline, and on declines, no fixed parameter set keeps it upright, so controller fragility rather than the adaptation method limits generalization to steeper terrain. Coupling the agent to closed-loop attitude feedback and validating on hardware are natural next steps.

VI. CONCLUSIONS

We showed that the CPG parameters producing stable quadruped locomotion shift with the terrain, so that a gait tuned for flat ground becomes unstable on an incline, and that across repeated transitions the terrain-matched optimum confers a durable survival advantage, so resuming stable walking calls for online adaptation. We proposed an event-triggered scheme in which a prediction-error monitor detects each terrain change and an adaptation method re-tunes the CPG within the same walking bout, and we compared grid search, Bayesian optimization, and an autoregressive active inference agent as that method. Online black-box search roughly doubles the fall rate by destabilizing the robot while probing candidate gaits, whereas the active inference agent selects from a learned model and avoids this harm; and once its goal prior encodes attitude stability rather than velocity, it significantly outperforms keeping the incumbent gait on a sustained transition, with the information-seeking term of the free energy essential to the gain. Selecting parameters from a model rather than by physical trial avoids destabilization, and encoding stability in the agent’s goal turns caution into improvement; making that improvement robust across recurring transitions, where the terrain optimum remains out of the model’s reach, is the main direction for future work.

APPENDIX

ACKNOWLEDGMENT

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